

ROOTWARD ACTIVE-PILE COALESCENCE ON A PATH: INTERFACE ADDITIVITY AND EXACT JUMP COUNTS

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ABSTRACT. We study a finite coalescing process on a rooted path. A discrete update selects one occupied nonroot site uniformly, moves its pile one edge toward the root, and erases multiplicity on contact. For every initial state $S = \{0 = s_0 < s_1 < \dots < s_r\}$, we prove that the expected number of updates to absorption is the additive interface functional

$$\mathbb{E}T_S = \sum_{i=1}^r h(s_{i-1}, s_i),$$

where h is the meeting-time mean of two ordered rate-one pure-death paths. The proof Poissonizes the active-pile scheduler, identifies the embedded jump chain exactly, and converts its jump count into the integral of the number of open initial interfaces. A first-passage recurrence determines h ; a ballot evaluation gives $h(m-1, m) = (2m-1)!!/(2m-2)!!$. Thus full occupancy on $\{0, \dots, n-1\}$ has mean

$$\sum_{m=1}^{n-1} \frac{(2m-1)!!}{(2m-2)!!} = \frac{4}{3\sqrt{\pi}} n^{3/2} + O(n^{1/2}).$$

We also give the complete support of the hitting-time law from every rooted state and the minimum-time mass from full occupancy. Exact rational controls corroborate all finite recurrences. Standard coalescing-walk, graphical, and first-passage machinery is treated as background; the owner search is bounded, and no novelty or external-release claim is made.

1. THE PROCESS AND THE SUBTRACTION BOUNDARY

Let S be a finite subset of $\mathbb{N}_0$ containing the root 0. If $S \neq \{0\}$, select $v \in S \setminus \{0\}$ uniformly and set

$$(1) \quad C_v(S) = (S \setminus \{v\}) \cup \{v-1\}.$$

If $v-1$ is already occupied, the two piles coalesce; the state records no multiplicity. Let T_S be the number of updates until $\{0\}$ is reached. This is an *active-pile* chain: uniform selection is among the current nonroot piles. Uniform selection from a fixed geometric set with lazy empty events gives a different discrete-time law.

Coalescing random walks and their voter-model duals form a classical interacting-particle framework; full occupancy, consensus time, and meeting-time comparison are established themes [2, 3, 5]. Graphical constructions and recurrence questions on infinite or rooted carriers are likewise standard [1]. Exact one-dimensional coalescence and reductions to simple-walk first passage also have a substantial history [4]. We assign zero contribution credit to coalescing-walk formalism, graphical arrows, voter duality, meeting/hitting comparisons, ballot identities, and central-binomial asymptotics.

The literal chain in (1) differs in three linked respects. Motion is deterministic and one-way, the scheduler is uniform among current piles, and T_S counts embedded updates rather than elapsed coalescence time. The residual question is whether this update count admits an exact formula from an arbitrary initial set. A bounded primary-source search found no statement of the theorem package below. That non-hit is not a novelty certificate, so external release remains on hold.

The internal subtraction is also strict. P114 already owns synchronous rooted-forest peeling, and P121 owns uniform adjacent coalescence together with its random-BST/Yule deletion history. P117's pointer-doubling and word-eroder exclusions remain binding, while P126 concerns synchronous length-increasing refinement. Rooted monotonicity, generic coalescence, adjacent-interface language, and random scheduling therefore receive no internal credit. The only claimed residual is the literal active-pile path chain, its general interface-additive jump count, and the consequences proved from that identity.

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2. THE FINITE LAW AND ITS SUPPORT

Write

$$\Phi(S) = \sum_{v \in S} v, \quad G_S(z) = \mathbb{E}[z^{T_S}].$$

Proposition 2.1 (absorption and PGF). *Every trajectory is absorbed after at most $\Phi(S)$ updates. Moreover,*

$$(2) \quad G_{\{0\}}(z) = 1, \quad G_S(z) = \frac{z}{|S| - 1} \sum_{v \in S \setminus \{0\}} G_{C_v(S)}(z).$$

Thus (2) is a finite acyclic recursion for the complete law.

Proof. If $v - 1 \notin S$, the update decreases Φ by one. If $v - 1 \in S$, the collision removes v and decreases Φ by v . The nonnegative integer potential therefore decreases strictly. Conditioning on the uniformly chosen first active site gives (2); potential descent makes the recursion acyclic. $\square$

The recursion has no internal gaps, even for a sparse initial state.

Theorem 2.2 (support from every rooted state). *If $S \neq \{0\}$, then*

$$(3) \quad \text{supp}(T_S) = \{\max S, \max S + 1, \dots, \Phi(S)\}.$$

Proof. Each update decreases Φ by at least one, so $T_S \leq \Phi(S)$. An update lowers the maximum occupied coordinate by at most one: moving the maximum either places it at its predecessor or removes it when that predecessor is occupied; a move below the maximum leaves it fixed. Hence $T_S \geq \max S$.

We prove that every intermediate value occurs by induction on $\Phi(S)$. Put $m = \max S$. If $m - 1 \notin S$, choose m first. The successor has maximum $m - 1$ and potential $\Phi(S) - 1$, so induction and the first update realize the whole interval $[m, \Phi(S)]$.

Suppose instead that $m - 1 \in S$. The case $m = 1$ is immediate. Choosing m first is a collision; induction supplies

$$(4) \quad [m, \Phi(S) - m + 1].$$

Let a be the bottom of the consecutive occupied run ending at m . If $a = 1$, moving a collides at the root; if $a > 1$, its predecessor is empty. Either move decreases the potential by one and leaves maximum m . This first choice and induction supply

$$(5) \quad [m + 1, \Phi(S)].$$

Since $m, m - 1 \in S$, we have $\Phi(S) \geq 2m - 1$; the integer intervals in (4)–(5) touch or overlap. Their union proves (3). Every constructed schedule has positive probability because each of its active choices does. $\square$

Let $S_n = \{0, 1, \dots, n - 1\}$ and $T_n = T_{S_n}$.

Corollary 2.3 (minimum-time mass). *For $n \geq 2$,*

$$(6) \quad \mathbb{P}(T_n = n - 1) = \frac{1}{(n - 1)!}.$$

Proof. A trajectory of length $n - 1$ must reduce the number of nonroot piles at every update, so every move must collide. Removing a site before its larger neighbor would leave a vacancy for that neighbor's next move. The unique all-collision order is therefore $n - 1, n - 2, \dots, 1$. The successive numbers of active sites are $n - 1, n - 2, \dots, 1$, giving (6). $\square$

3. POISSON CLOCKS AND INTERFACE ADDITIVITY

The clock construction must preserve the discrete scheduler, not merely the set of possible moves. Put an independent rate-one Poisson clock at each positive site. A ring at an occupied site triggers (1); a ring at an empty site is ignored, and the root has no clock. If the current state has k nonroot piles, the next effective ring is the minimum of k independent rate-one exponentials. Its site is uniform among those k sites. Hence the embedded effective-jump chain is exactly the chain of section 1.

For $0 \leq a \leq b$, define $h(a, b)$ by

$$(7) \quad h(a, a) = 0,$$

$$(8) \quad h(0, b) = b,$$

$$(9) \quad h(a, b) = \frac{1}{2} + \frac{h(a - 1, b) + h(a, b - 1)}{2} \quad (0 < a < b).$$

This triangular recursion uniquely defines nonnegative rational values.

Theorem 3.1 (interface-additive update count). *Let $S = \{0 = s_0 < s_1 < \dots < s_r\}$. Then*

$$(10) \quad \mathbb{E}[T_S] = \sum_{i=1}^r h(s_{i-1}, s_i).$$

The empty sum for $S = \{0\}$ equals zero.

Proof. Label the initial piles $0, 1, \dots, r$ from left to right and follow each label through the site-clock arrows. Labels that meet share one path thereafter. Rootward nearest-neighbor paths preserve order: they can coalesce but cannot cross.

For $1 \leq i \leq r$, let τ_i be the first meeting time of the paths started at s_{i-1} and s_i . At time t , consecutive initial labels belong to the same current pile exactly when their interface has closed. The current piles are therefore the consecutive label blocks separated by open interfaces. One block contains the root, so the number N_t of nonroot piles satisfies the pathwise identity

$$(11) \quad N_t = \sum_{i=1}^r \mathbf{1}_{\{\tau_i > t\}}.$$

Let J_t count effective updates by continuous time t . Conditional on the current state, its total rate is N_t ; thus $J_t - \int_0^t N_u du$ is the compensated jump-count martingale. By [proposition 2.1](#), $J_\infty \leq \Phi(S)$, so monotone convergence and Tonelli's theorem give

$$(12) \quad \begin{aligned} \mathbb{E}T_S &= \mathbb{E}J_\infty = \mathbb{E} \int_0^\infty N_t dt \\ &= \sum_{i=1}^r \mathbb{E} \int_0^\infty \mathbf{1}_{\{\tau_i > t\}} dt = \sum_{i=1}^r \mathbb{E}\tau_i. \end{aligned}$$

It remains to identify one interface marginal. Before two paths at $0 < a < b$ meet, they occupy distinct sites and read independent rate-one clocks. The next event takes mean time $1/2$; either the lower path moves to $a - 1$ or the upper path moves to $b - 1$, each with probability $1/2$. The strong Markov property supplies fresh exponentials even when a site was visited earlier. At equality the interface is closed, while a lower path at zero is fixed and the upper path needs b mean-one moves. First-event conditioning is exactly (7)–(9), so $\mathbb{E}\tau_i = h(s_{i-1}, s_i)$. Substitution into (12) proves (10). $\square$

Remark 3.2. The interface lifetimes are generally dependent. Equation (10) uses only the pathwise count (11), Tonelli's theorem, and their individual means.

4. A BALLOT EVALUATION AND FULL OCCUPANCY

The recurrence for h is already an exact finite algorithm. Consecutive starts have a closed form.

Lemma 4.1 (adjacent first passage). *For every $m \geq 1$,*

$$(13) \quad h(m-1, m) = \frac{(2m-1)!!}{(2m-2)!!} = \frac{2m}{4^m} \binom{2m}{m}.$$

Equivalently, this value is $\mathbb{E}|W_{2m}|$ for a length- $2m$ simple symmetric walk W .

Proof. The case $m = 1$ is (8). Suppose $m \geq 2$ and put $p = m - 1$. Superpose the two clocks until the paths meet or the lower path reaches zero. Each inter-event time has mean $1/2$, and the event type is a fair coin. Call a lower-path move an up-step and an upper-path move a down-step. The gap starts at one; a down-step at gap one closes the interface, while the p th up-step puts the lower path at zero.

If meeting occurs after j up-steps and $j + 1$ down-steps, the event word is a Catalan first-passage word. There are

$$A_j = \frac{1}{j+1} \binom{2j}{j}, \quad 0 \leq j < p,$$

such words. If the p th up-step occurs after q down-steps, reflection at the first forbidden prefix gives

$$B_{p,q} = \binom{p+q-1}{q} - \binom{p+q-1}{q-1}, \quad 0 \leq q < p,$$

where the second binomial coefficient is zero at $q = 0$. On this second exit, the upper path is $p + 1 - q$ steps from zero. Consequently

$$(14) \quad h(m-1, m) = \sum_{j=0}^{p-1} \frac{A_j}{2^{2j+1}} \frac{2j+1}{2} + \sum_{q=0}^{p-1} \frac{B_{p,q}}{2^{p+q}} \left(\frac{p+q}{2} + p + 1 - q \right).$$

The first parenthesis is elapsed mean time; the second also includes the one-clock residual time after the lower path reaches zero.

For completeness, use

$$B_{p,q} = \frac{p-q}{p+q} \binom{p+q}{q}$$

in (14). Pascal's identity, with the $q = 0$ and $q = p - 1$ terms kept as boundaries, gives for R_m equal to that sum

$$(15) \quad R_1 = 1, \quad 2mR_{m+1} = (2m+1)R_m.$$

The double-factorial ratio in (13) has the same initial value and recurrence. The central-binomial form follows by cancelling even and odd factors. Finally, pairing the positive and negative endpoints in the binomial law of W_{2m} gives $\mathbb{E}|W_{2m}| = 2m \binom{2m}{m} / 4^m$. $\square$

Theorem 4.2 (full-start mean). *For $n \geq 1$,*

$$(16) \quad \mathbb{E}T_n = \sum_{m=1}^{n-1} \frac{(2m-1)!!}{(2m-2)!!}.$$

As $n \rightarrow \infty$,

$$(17) \quad \mathbb{E}T_n = \frac{4}{3\sqrt{\pi}} n^{3/2} + O(n^{1/2}).$$

Proof. The consecutive interfaces of S_n start at $(m-1, m)$ for $1 \leq m < n$. Equations (10) and (13) give (16). The standard estimate

$$\binom{2m}{m} = \frac{4^m}{\sqrt{\pi m}} (1 + O(m^{-1}))$$

makes the m th summand $2\sqrt{m/\pi} + O(m^{-1/2})$. Summing and comparing $\sum_{m < n} \sqrt{m}$ with its integral gives (17). $\square$

5. EXACT CONTROL AND SCOPE

The paper-local verifier constructs (1) literally and uses integer arithmetic and `fractions.Fraction` only. It compares the Bellman expectation with the independent interface sum for every rooted subset through $n = 14$ (16,383 states), audits every rooted hitting-time law through $n = 11$ (2,047 states), checks the full pair recurrence triangle $0 \leq a \leq b \leq 80$, and evaluates the Catalan/ballot sum independently for $m \leq 80$. It also checks the full-start law and [corollary 2.3](#). The deterministic run executes 506,663 assertions and its stdout is frozen byte for byte. These finite controls expose boundary, scheduler, and indexing errors; they are not substitutes for the proofs.

One additional upper-endpoint pattern is retained in the program under the literal label `PILOT_ONLY`. It is not a theorem, does not appear in the abstract or contribution list, and supplies no promotion credit.

The scope is deliberately narrow. The state is a set, the root never rings, every positive pile has the same active rate, and motion is deterministically one edge toward zero. Retaining mass labels, assigning clocks permanently to original particles, sampling geometric sites with lazy holds, using unbiased motion, or replacing the path by a tree changes the process. No claim here transfers to those variants.

6. CONCLUSION

The active-pile chain has an exact arbitrary-state statistic: its expected number of embedded updates is the sum of the meeting-time means of consecutive initial interfaces. The site-clock construction proves scheduler equivalence, while the compensator converts elapsed interface lifetimes back to the discrete update count. A ballot calculation then yields the double-factorial full-start mean and its $n^{3/2}$ scale. The finite PGF and support theorem provide a second description of the hitting law.

The result is path-specific and does not assert independent interfaces. Generic coalescing-walk and first-passage methods have been subtracted, and the source search remains bounded. Accordingly, novelty, priority, posting, submission, and external release remain undecided.

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